

Zadanie 25

Oblicz granicę $\lim_{n \rightarrow \infty} (n - \sqrt{n+3})$.

① Obliczamy granicę.

$$\lim_{n \rightarrow \infty} (n - \sqrt{n+3}) = \lim_{n \rightarrow \infty} n \left(1 - \sqrt{\frac{1}{n} + \frac{3}{n^2}}\right) = \lim_{n \rightarrow \infty} n (1 - 0) = \lim_{n \rightarrow \infty} n = \infty$$

$$\lim_{n \rightarrow \infty} \sqrt{n+3} = \lim_{n \rightarrow \infty} \sqrt{n^2 \left(\frac{1}{n} + \frac{3}{n^2}\right)} = \lim_{n \rightarrow \infty} n \sqrt{\underbrace{\frac{1}{n}}_0 + \underbrace{\frac{3}{n^2}}_0} = 0.$$

$$\text{Odp.: } \lim_{n \rightarrow \infty} (n - \sqrt{n+3}) = \infty.$$